

Nizam-e-Qunoon: A Proposal for Pakistan's First Intelligent Legal Infrastructure

1. Introduction and Project Vision

1.1 The Imperative for Digital Justice in Pakistan

The Islamic Republic of Pakistan stands at a critical juncture in its legal history, characterized by a stark paradox between constitutional guarantees and the lived reality of its citizens. While Article 10-A of the Constitution of Pakistan guarantees the right to a fair trial and due process, the structural machinery required to deliver this right is overwhelmed. Recent judicial statistics reveal a staggering backlog exceeding 2.3 million cases pending across the judiciary.¹ This systemic congestion is not merely an administrative failure; it represents a profound denial of justice for millions. The situation is further exacerbated by the "literacy barrier." With a national literacy rate stagnating around 60.6%, and significantly lower in rural areas², the text-heavy, archaic lexicon of the legal system renders it unintelligible to nearly half the population.

In this context, *Nizam-e-Qunoon* (System of Law) is proposed not merely as a mobile application, but as a foundational shift in how legal services are accessed, understood, and delivered in Pakistan. The project envisions the creation of Pakistan's first intelligent legal infrastructure. Drawing inspiration from how platforms like Facebook democratized social connectivity or YouTube democratized media distribution, Nizam-e-Qunoon aims to democratize legal access. By leveraging the confluence of modern software engineering—specifically Mobile-First architecture—and advanced Artificial Intelligence (AI) techniques like Retrieval-Augmented Generation (RAG) and Parameter-Efficient Fine-Tuning (PEFT), the platform will serve as a comprehensive, bilingual legal ecosystem.

The current legal technology landscape in Pakistan is fragmented and largely static. Government initiatives such as the *Pakistan Code* website³ or the *Pakistan Citizen Portal*⁴ serve as digital repositories or grievance lodging mechanisms, respectively. They lack "intelligence"—the ability to interpret a user's natural language query, map it to the relevant statutes, and provide actionable guidance. A citizen dealing with a landlord dispute cannot simply type "my landlord is evicting me without notice" into existing portals and receive a coherent explanation of the *West Pakistan Urban Rent Restriction Ordinance, 1959*. Instead, they are forced to navigate complex drop-down menus or hire expensive counsel. Nizam-e-Qunoon addresses this gap by building an AI-powered legal assistant that understands context, applies relevant laws to user scenarios, and generates court-ready

documents [User Query].

1.2 Alignment with National Policy and SDGs

This proposal is strategically aligned with the "National AI Policy 2025," approved by the Federal Cabinet, which targets the training of one million AI professionals and the deployment of AI in governance to transform Pakistan into a knowledge-based economy.⁵ The policy explicitly calls for the establishment of an "AI Innovation Ecosystem" and the "Transformation and Evolution" of public sectors, including governance and citizen services.⁷ Nizam-e-Qunoon directly contributes to these national objectives by building domestic AI capabilities in the high-stakes domain of law.

Furthermore, the project supports the United Nations Sustainable Development Goal (SDG) 16: "Peace, Justice, and Strong Institutions," specifically Target 16.3, which aims to promote the rule of law and ensure equal access to justice for all.² By reducing the information asymmetry between legal elites and the general public, Nizam-e-Qunoon serves as a technological enabler for statutory rights.

1.3 Project Scope and Core Objectives

The scope of this Final Year Project (FYP) encompasses the end-to-end development of a functional Minimum Viable Product (MVP) that demonstrates the efficacy of Generative AI in the Pakistani legal context. The primary objectives are:

1. **Democratization of Legal Knowledge:** To develop a mobile-first platform that provides instant, free access to federal and provincial laws, case law, and regulations, accessible even in offline environments common in Pakistan [User Query].
2. **Context-Aware Legal Intelligence:** To engineer a hybrid RAG system that combines keyword search (BM25) with semantic vector search (Dense Retrieval), enabling the system to understand the *intent* behind a user's query rather than just matching words.
3. **Bilingual Accessibility (Urdu/English):** To bridge the linguistic divide by fine-tuning models on the *LEGAL-UQA* dataset and other Urdu corpora⁸, allowing users to query in their native language and receive legally accurate responses.
4. **Automated Document Generation:** To implement a system that generates legally sound documents (e.g., Rent Agreements, Affidavits) using AI-driven template completion and LaTeX rendering, reducing the reliance on manual drafting.
5. **Ethical and Verifiable AI:** To mitigate the risk of "hallucination" (AI fabricating laws) by grounding all responses in a verifiable database of Pakistani statutes and implementing strict citation protocols.⁹

2. Problem Diagnosis and Background Analysis

2.1 The Crisis of Accessibility and Efficiency

The Pakistani legal system is plagued by structural inefficiencies that make justice a luxury good. The backlog of 2.3 million cases is a symptom of a deeper malaise: the reliance on manual, paper-based processes and the scarcity of efficient research tools.¹ Lawyers and judges spend disproportionate amounts of time searching for precedents in physical law journals or disconnected databases. This inefficiency trickles down to the client in the form of higher fees and delayed justice.

For the common citizen, the "Black Box" of the law is impenetrable. The legal framework is a complex amalgam of British Common Law, Islamic *Shariat*, and post-independence statutes. A citizen seeking to understand their rights regarding inheritance, for instance, must navigate the *Muslim Family Laws Ordinance, 1961*, various interpretations by the Federal Shariat Court, and secular property laws. The lack of a unified, intelligent interface means that "ignorance of the law" is not just an excuse, but a forced reality for the majority.

2.2 The Limitations of Existing Solutions

Current digital legal solutions in Pakistan fail to meet the needs of the modern user.

- **Government Repositories:** The *Pakistan Code* ³ is an authoritative source but lacks semantic search. If a user searches for "firing an employee," they may find no results because the legal term is "termination of service" or "dismissal." These systems require the user to *already know* the law to find the law.
- **Grievance Apps:** The *Pakistan Citizen Portal* ⁴ allows users to lodge complaints but does not provide legal guidance. Reviews indicate issues with user experience and a lack of transparency in how complaints are processed. It is a routing mechanism, not an advisory one.
- **Private Sector Competitors:** Startups like *Lexa* and *Wusuq* ¹⁰ have begun to enter the market. *Lexa* offers a chatbot and judgment database, but the market is still underserved in terms of *generative* document capabilities and deep semantic understanding of Urdu queries. Furthermore, many of these tools rely on generic LLM wrappers that are prone to hallucinating US laws instead of Pakistani statutes due to a lack of domain-specific fine-tuning.

2.3 The AI "Hallucination" Risk in Law

A critical challenge in deploying Generative AI in law is the phenomenon of hallucination—where an AI model confidently asserts false information. In the US case of *Mata v. Avianca*, lawyers were sanctioned for submitting a brief containing non-existent cases generated by ChatGPT.¹² In the Pakistani context, an AI might erroneously cite the Indian Penal Code instead of the Pakistan Penal Code due to the similarity in their colonial origins. Standard "off-the-shelf" LLMs (like GPT-4 or standard Llama 3) have high hallucination rates in specialized domains, sometimes estimated between 58% and 80% for legal tasks.⁹ This necessitates a rigorous architectural approach: Retrieval-Augmented Generation (RAG). Unlike a standard chatbot that relies on internal memory, a RAG system retrieves specific

documents from a trusted database and forces the AI to answer only using that retrieved context. Research indicates that RAG significantly improves faithfulness and attribution accuracy in legal domains.⁹

2.4 The Urdu Natural Language Processing Gap

Developing for Pakistan requires robust Urdu support. However, Urdu is a "low-resource" language in the NLP world compared to English or Chinese. Standard multilingual models often struggle with the complexity of the Nastaliq script and the specific legal vocabulary of Urdu (which borrows heavily from Persian and Arabic).

Recent academic work, such as the release of the LEGAL-UQA dataset (derived from the Constitution of Pakistan), highlights the gap. This dataset contains only ~600 QA pairs⁸, which is insufficient for training a robust model from scratch but valuable for fine-tuning. Existing libraries like UrduHack¹⁴ and LughaatNLP¹⁵ provide essential tools for tokenization and normalization, but a comprehensive "Legal Urdu" model remains a gap that Nizam-e-Qunoon aims to fill.

3. Technical Architecture and System Design

The architecture of Nizam-e-Qunoon is designed to be **hybrid, modular, and scalable**. It avoids the pitfalls of monolithic designs by separating the AI inference engine from the business logic, ensuring that the heavy computational load of the LLM does not degrade the user experience of the mobile app.

3.1 High-Level Architecture Components

The system is composed of four primary layers:

1. **The Client Layer (Flutter Mobile App):** A cross-platform interface ensuring reach across Android and iOS devices.
2. **The Application Layer (FastAPI Backend):** A high-performance, asynchronous Python web framework that orchestrates API requests, user authentication, and business logic.
3. **The Intelligence Layer (AI & RAG Pipeline):** The core "brain" consisting of the fine-tuned Llama 3.1 model, the retrieval systems, and the document generation engine.
4. **The Data Layer (Hybrid Storage):** A combination of PostgreSQL for relational data (users, structured laws) and Qdrant/Weaviate for vector embeddings.

3.2 The RAG Pipeline: A Multi-Stage Retrieval Strategy

To ensure maximum accuracy and minimize hallucinations, we reject the naive "single-vector-search" approach. Instead, we propose a sophisticated **Multi-Stage Retrieval Pipeline** [User Query].

Stage	Technology	Function	Justification
1. Sparse Retrieval	BM25 (Best Matching 25)	Keyword-based search.	Legal research often requires exact matches (e.g., searching for "Section 302"). Vector search often misses exact numbers or specific legal phrasing. BM25 ensures these are captured. ¹³
2. Dense Retrieval	Sentence Transformers (Fine-tuned)	Semantic/Contextual search.	Captures the <i>meaning</i> of a query. If a user asks about "firing," this layer retrieves laws related to "termination," mapping synonyms in the vector space. ¹⁶
3. Reranking	Cross-Encoder (Bert-based)	Relevance scoring.	The top ~50 results from stages 1 and 2 are passed to a Cross-Encoder. This model reads the query and the document pair together to output a high-fidelity relevance score, filtering out noise before it reaches the LLM. ¹³

Why this matters: In a legal context, retrieving the *wrong* law is worse than retrieving *no* law. The Cross-Encoder acts as a strict gatekeeper, ensuring that the LLM is only fed the most

pertinent statutory context.

3.3 The Generative Engine: Fine-Tuning Llama 3.1

We will utilize the **Llama 3.1 (8B Parameters)** model as our base. This model offers a balance between reasoning capability and computational feasibility.

- **Domain Adaptation:** Generic models speak "internet English." To make the model speak "Pakistani Law," we must fine-tune it. We will use a curated corpus comprising the Constitution, the Pakistan Penal Code (PPC), the Code of Criminal Procedure (CrPC), and anonymized Supreme Court judgments.
- **Parameter-Efficient Fine-Tuning (PEFT) via LoRA:** Full fine-tuning of an 8B model requires massive GPU resources (approx. 120GB+ VRAM). We will employ **Low-Rank Adaptation (LoRA)**.¹⁷ LoRA freezes the pre-trained model weights and injects trainable rank-decomposition matrices into each layer.
 - *Mechanism:* Instead of updating the massive weight matrix W , we update two smaller matrices A and B such that $\Delta W = BA$.
 - *Benefit:* This reduces the number of trainable parameters by 10,000x and GPU memory requirements to <16GB¹⁹, making it feasible to train on consumer hardware (e.g., NVIDIA RTX 3090 or Google Colab T4) suitable for an FYP budget.
- **Quantization (QLoRA):** To further optimize for deployment, we will use QLoRA (Quantized LoRA), loading the base model in 4-bit precision. This allows the model to run on edge servers or lower-cost cloud instances without significant loss in reasoning quality.²¹

3.4 Automated Document Generation (LaTeX + AI)

While many legal apps offer simple "fill-in-the-blank" forms, Nizam-e-Qunoon introduces **Intelligent Drafting**.

1. **Contextual Data Extraction:** The AI engages the user in a dialogue to understand the specific requirements (e.g., "Is this rent agreement for a commercial shop or a house?").
2. **Logic Injection:** Based on the answers, the AI selects the appropriate clauses. For example, if the property is commercial, it inserts clauses related to commercial tax liabilities which are absent in residential agreements.
3. **LaTeX Rendering:** The system uses **LaTeX** for the final output. Unlike HTML-to-PDF converters, LaTeX ensures professional typesetting, precise pagination, and court-acceptable formatting [User Query]. The dynamic content from the AI is injected into LaTeX templates to generate a high-quality PDF.

3.5 Mobile Architecture: Offline-First

Given Pakistan's intermittent internet connectivity, particularly in rural districts, an **Offline-First** approach is critical.

- **Local Caching:** We will use local databases (like *Isar* or *SQLite* in Flutter) to store the full

text of major codes (Constitution, PPC).

- **Sync Protocol:** A synchronization engine will check for updates when connectivity is restored. This is vital for laws that change frequently, such as tax codes or traffic fines.
- **On-Device Vector Search:** For basic queries, we will explore lightweight, on-device vector stores (e.g., *ObjectBox* or *TFLite*) to allow semantic search even without server access.

4. Methodology and Implementation Roadmap

The development of Nizam-e-Qunoon will follow an iterative, Agile methodology. This ensures that a functional prototype is available early for testing, with features added in concentric circles of complexity.

4.1 Data Acquisition and Engineering

- **Scraping Pipeline:** We will utilize **GitHub Actions** to automate the daily scraping of legal sources.
 - *Sources:* Pakistan Code (Federal Laws), Provincial Assembly websites, and Supreme Court websites.
 - *Change Detection:* The pipeline will implement hashing algorithms to detect changes in the source text. If a law is amended, the system detects the diff, updates the database, and triggers a re-indexing of the vector store.
- **OCR and Digitization:** A significant challenge is that many older judgments are scanned images. We will implement an OCR pipeline using **Tesseract** for English and **UTRNet**⁸ for Urdu text. UTRNet is specifically optimized for the complex layouts of Urdu legal documents.
- **Anonymization:** To train our models on case law without violating privacy, we will use Named Entity Recognition (NER) (via *SpaCy* or *LughaatNLP*) to automatically redact names, CNIC numbers, and addresses from the training corpus.¹⁵

4.2 Development Phases (Gantt Chart Strategy)

Based on standard FYP timelines²², the project is divided into three distinct phases over a 6-month period.

Phase 1: Foundation & Infrastructure (Months 1-2)

- **Goal:** Establish the data layer and basic retrieval system.
- **Tasks:**
 - Develop web scrapers and populate the PostgreSQL database.
 - Set up the Qdrant vector database and generate initial embeddings using sentence-transformers.
 - Build the FastAPI shell with basic CRUD endpoints.

- Develop the sparse retrieval (BM25) module.
- **Deliverable:** A backend API that can return correct legal sections based on keyword search.

Phase 2: Intelligence & Hybrid RAG (Months 3-4)

- **Goal:** Implement the "Brain" of the system.
- **Tasks:**
 - Fine-tune Llama 3.1 8B using LoRA on the prepared legal dataset.
 - Integrate the Cross-Encoder reranker to combine Sparse and Dense results.
 - Implement the LaTeX document generation engine with 5-10 core templates (Rent Agreement, Divorce Deed, Legal Notice).
 - Develop the "Scenario Analysis" feature where users describe a situation and get legal reasoning.
- **Deliverable:** A functioning RAG pipeline and document generator.

Phase 3: Client Application & Production Readiness (Months 5-6)

- **Goal:** Mobile app and deployment.
- **Tasks:**
 - Develop the Flutter application with offline capabilities.
 - Implement bilingual UI (Urdu/English) toggles.
 - Containerize the entire stack using Docker and orchestrate with Kubernetes for scalability.
 - Conduct User Acceptance Testing (UAT) with law students.
- **Deliverable:** Deployment-ready mobile app and final thesis report.

4.3 Evaluation Metrics

To ensure academic rigor, the system will be evaluated using quantitative metrics:

- **Retrieval Accuracy:** We will measure *Recall@K* and *MRR (Mean Reciprocal Rank)* to assess how often the correct law is found in the top 5 results.
- **Generation Faithfulness:** We will use the **RAGAS (Retrieval Augmented Generation Assessment)** framework.²⁴ This framework uses a "Judge LLM" (like GPT-4) to score our model's responses on:
 - *Faithfulness:* Does the answer contradict the retrieved context?
 - *Answer Relevance:* Does the answer actually address the user's question?
 - *Context Precision:* Was the retrieved context actually useful?
- **Hallucination Rate:** We will conduct "Red Teaming" exercises to try and force the model to invent laws, measuring the failure rate before and after implementing the RAG pipeline.

5. Ethical Considerations, Risks, and Legal Compliance

5.1 The "Unauthorized Practice of Law" Risk

A primary risk for any "Robot Lawyer" is crossing the line from providing *legal information* to providing *legal advice*. Under the *Legal Practitioners and Bar Councils Act* of Pakistan, only licensed advocates may practice law.

- **Mitigation Strategy:** Nizam-e-Qunoon will implement strict, prominent disclaimers.²⁶ Every interaction will begin with a statement clarifying that the AI provides educational information and is not a substitute for professional counsel. The output will be framed as "The law states X" rather than "You should do X."

5.2 Data Privacy and Security

Users may enter sensitive personal information (e.g., details of a crime or family dispute) into the app.

- **Encryption:** All data in transit will be encrypted via TLS 1.3. Data at rest in the PostgreSQL database will be encrypted using AES-256 standards.
- **Data Minimization:** We will implement an "Anonymous Mode" where user queries are processed in volatile memory (RAM) and never logged to the database, ensuring zero retention of sensitive queries.
- **Regulatory Compliance:** The system design anticipates the enactment of the *Personal Data Protection Bill* of Pakistan, ensuring mechanisms for users to request data deletion (Right to be Forgotten).²⁸

5.3 Algorithmic Bias and Fairness

AI models trained on historical data can inherit biases. In the legal context, this could manifest as biased language regarding gender or minority groups found in older judgments.

- **Mitigation:** We will rigorously curate the fine-tuning dataset to exclude discriminatory language. We will also implement a "Safety Layer" (using libraries like Guardrails AI) to filter out any output that violates ethical guidelines or promotes hate speech.

6. Business Model and Commercial Viability

While initially an academic project, Nizam-e-Qunoon is designed for commercial sustainability. The tech startup landscape in Pakistan, particularly in Fintech and Healthtech, has shown resilience with startups raising \$36.6 million in 2025.²⁹ LegalTech is the next frontier.

6.1 Target Market

1. **General Public (B2C):** The 100+ million mobile users in Pakistan who need basic legal information.

2. **Law Students & Junior Advocates (B2B):** A subscription market of thousands of professionals who cannot afford expensive legacy databases but need reliable research tools.
3. **SMEs (B2B):** Small businesses needing automated contract drafting and regulatory updates.

6.2 Revenue Strategy: The Freemium Model

- **Free Tier:** Access to the Constitution, Federal Laws, and basic keyword search. This drives user acquisition and serves the social mission of democratization.
- **Premium Tier:** Access to the AI Assistant (Unlimited RAG queries), Automated Document Generation (Rent agreements, etc.), and Offline Mode.
- **Enterprise API:** Providing our "Legal Intelligence API" to other platforms (e.g., Real Estate portals needing to check property laws) [User Query].

6.3 Scalability and Cloud Architecture

To ensure the system handles national-scale traffic:

- **Serverless Inference:** To manage the high cost of GPU hosting, we will utilize serverless GPU platforms (like Modal or RunPod). This ensures we only pay for the seconds the AI is actually generating text, rather than paying for idle servers.
- **Containerization:** The entire application is dockerized. Using Kubernetes, we can horizontally scale the FastAPI backend nodes to handle thousands of concurrent requests during peak times.
- **CI/CD Pipeline:** The automated data pipeline ensures the system is "self-healing" and "self-updating." As soon as a new law is published on the Gazette website, the GitHub Action scraper triggers, updates the vector store, and pushes a notification to the user app, ensuring zero latency in legal currency.

7. Conclusion

Nizam-e-Qunoon represents more than just a software application; it is a blueprint for the modernization of Pakistan's legal infrastructure. By bridging the gap between the rigid, text-heavy world of law and the fluid, accessible world of mobile technology, it addresses the fundamental crisis of accessibility that plagues the justice system.

This proposal outlines a rigorous technical path to achieving this vision. We move beyond the hype of AI to implement concrete, verifiable engineering solutions: **RAG** to solve hallucination, **LoRA** to solve computational constraints, and **Offline-First Flutter Architecture** to solve connectivity issues. By grounding the system in the specific realities of Pakistan—its languages, its laws, and its infrastructure challenges—Nizam-e-Qunoon aims to be not just an academic success, but a pillar of digital justice for the nation.

Detailed Technical Report

4.1 System Architecture Deep Dive

4.1.1 The Backend Ecosystem (FastAPI & Python)

The choice of **FastAPI** is deliberate. Unlike Django (which is monolithic) or Flask (which lacks native async support), FastAPI is built on *Starlette* and *Pydantic*, offering high-performance asynchronous request handling. This is crucial for an AI application where inference can take several seconds; async handling prevents the server from blocking other user requests while waiting for the GPU to respond.

- **Integration:** FastAPI's native support for Python allows seamless integration with PyTorch and LangChain libraries within the same codebase, reducing latency compared to a microservices architecture where the API and AI are in different languages.

4.1.2 Database Schema and Vector Store

The system utilizes a dual-database approach:

1. **PostgreSQL (Relational):**
 - *Tables:* Users, Subscriptions, DocumentHistory, Statutes (Text, metadata, amendment dates).
 - *Why:* ACID compliance is necessary for user data and subscription management.
2. **Qdrant (Vector):**
 - *Collections:* Statutes_Vector, CaseLaw_Vector.
 - *Indexing:* We use **HNSW (Hierarchical Navigable Small World)** graphs for the index. This allows for approximate nearest neighbor search which is exponentially faster than brute-force search.
 - *Embeddings:* We will use the BAAI/bge-m3 model for generating embeddings. This model supports multi-lingual input and long context lengths (up to 8192 tokens), making it superior to standard BERT models for legal texts.¹³

4.1.3 The AI Fine-Tuning Methodology (LoRA)

Fine-tuning a Large Language Model (LLM) is the process of updating its weights to better perform on a specific task.

- **The Challenge:** Updating all 8 billion parameters of Llama 3.1 requires storing the model, the gradients, and the optimizer states in GPU memory. For an 8B model in 16-bit precision, the model alone is ~16GB. The optimizer states (AdamW) take another ~32GB. Gradients take another ~16GB. This totals >64GB VRAM, requiring A100 GPUs which are prohibitively expensive (\$2-\$4/hour).

- **The Solution (LoRA):** We inject low-rank matrices into the attention layers of the transformer. We freeze the main weights (W_O) and only train the delta ($\Delta W = B \times A$).
 - A and B are much smaller (Rank $r=16$ or 32).
 - This reduces trainable parameters to $<1\%$ of the total.
 - **Result:** We can fine-tune on a single 24GB GPU (RTX 3090/4090) or even a 16GB T4 (using 4-bit quantization/QLoRA).¹⁹
 - **Data Strategy:** The training data will be formatted as instruction-response pairs:
 - *Instruction:* "Explain the punishment for theft under Pakistan Penal Code."
 - *Response:* "According to Section 378 of the PPC..."
 - This teaches the model the *structure* of a legal answer.

4.2 Application Features and User Experience

4.2.1 Bilingual Interface (Urdu Localization)

The interface will not rely on simple machine translation APIs, which often mangle legal terms. Instead, the UI will be fully localized with hard-coded legal terminology verified by human experts.

For the AI chat, we will employ a Translation Layer:

- *Input:* User types in Urdu.
- *Processing:* The query is translated to English using a specialized legal-translation model (fine-tuned Helsinki-NLP or NLLB).
- *Retrieval:* Search is performed in the English database (where the laws are most robust).
- *Generation:* The AI generates the answer in English.
- *Output:* The answer is translated back to Urdu for the user.
- *Rationale:* While we are building Urdu datasets, the English legal corpus is vastly larger. This "Cross-Lingual Information Retrieval" (CLIR) approach leverages the rich English data to serve Urdu users.

4.2.2 Interactive Document Automation

The "Document Generator" uses a template-slot approach.

- **Template:** A standard Rent Agreement in LaTeX format with variables like $\{\{LANDLORD_NAME\}\}$, $\{\{RENT_AMOUNT\}\}$, $\{\{PROPERTY_ADDRESS\}\}$.
- **AI Interview:** The LLM acts as the interviewer. It is given a prompt: "Ask the user for the necessary details to fill these variables. If the user is unsure about the 'Security Deposit' clause, explain the standard practice in Pakistan."
- **Validation:** Once the user provides data, the AI validates it (e.g., ensuring the CNIC format is correct).
- **Generation:** The variables are passed to a LaTeX compiler on the backend, generating a PDF.

5. Market Analysis and Commercialization

5.1 Competitor Landscape

- **Local:**
 - *Pakistan Law Site*: Legacy, web-only, expensive, no AI.
 - *Pindi Law*: Static mobile app, outdated data, ad-heavy.
 - *Lexa*: Emerging AI competitor.¹¹ Our advantage is the *Generative Document* feature and the *Offline-First* architecture, which Lexa lacks.
- **International:**
 - *Harvey / Casetext*: Enterprise-focused, prohibitively expensive (\$ thousands/month), zero knowledge of Pakistani procedural law.

5.2 Funding and Growth

The project is well-positioned to secure pre-seed funding. With the *National AI Fund* launching under the 2025 Policy ⁷, grants are available for civic-tech AI projects. Additionally, the recent trend of venture capital flowing into Pakistani startups (e.g., *Abhi*, *SadaPay*, and the recent \$74M funding surge in 2025 ³¹) indicates investor appetite for scalable, local software solutions.

6. Project Timeline (Gantt Chart)

Month	Phase	Key Milestones	Deliverables
Month 1	Inception	• Requirement Analysis • DB Schema Design • Scraper Development	• SRD Document • Populated PostgreSQL DB
Month 2	Core Backend	• FastAPI Setup • Vector DB Integration • BM25 Implementation	• Search API (Keyword) • Embedding Pipeline

Month 3	AI Development	• Dataset Cleaning (OCR/NER) • LoRA Fine-Tuning • RAG Pipeline V1	• Fine-tuned Llama 3.1 Model • Chat API Endpoint
Month 4	Doc Gen & Logic	• LaTeX Template Creation • Doc Gen Engine • Cross-Encoder Reranking	• Document Generation API • Improved Search Accuracy
Month 5	Frontend (App)	• Flutter UI/UX • API Integration • Local Caching (SQLite)	• Beta Mobile App (Android) • Offline Mode
Month 6	QA & Finalization	• User Testing (UAT) • Red Teaming (Hallucination Checks) • Final Deployment	• Production Release • Final FYP Report

7. Conclusion

Nizam-e-Qunoon is a technically rigorous, socially relevant, and commercially viable project. It addresses the "Black Box" problem of the Pakistani legal system by illuminating it with the light of Artificial Intelligence. By meticulously engineering a solution that respects the constraints of the local environment (low connectivity, low resources, diverse languages) while leveraging state-of-the-art global technology (Llama 3, RAG, LoRA), this project aims to set a new standard for LegalTech in the developing world. It fulfills the academic requirements of a Final Year Project while laying the groundwork for a platform that could one day serve millions of Pakistani citizens.

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